2) Attach a separate document explaining how eigen values and eigen vectors are calculated from scratch, including the characterstic equation and the steps to find the eigen vectors for each eigenvalue. Provide a simple example to demonstrate the process.

<u>Steps</u>

Phase 1 ⇒ preparing data

⇒ Calculate mean

⇒ Center the Data

Phase 2 ⇒ compute the covariance Matrix (Σ)

⇒ Var(x)

⇒ Var(y)

⋮

⇒ cov(x,y)

⋮

Phase 3 ⇒ compute Eigenvalue (λ)

$$\det(\Sigma - \lambda I) = 0$$

Phase 4 ⇒ Find Eigenvectors (v)

$$(\Sigma - \lambda I)v = 0$$

Phase 5 ⇒ compute final diamension Transformation

⇒ Normalize Vector

⇒ Find new values

<u>Example</u> Lets compute 1D computation of 2D Matrix

$$\begin{bmatrix} 2 & 4 \\ 4 & 2 \\ 6 & 8 \\ 8 & 6 \end{bmatrix}$$

We will calculate eigenvalues and eigenvector.

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Pase 1

$$\begin{array}{cc} x & y \\ \end{array}$$
$$\begin{bmatrix} 2 & 4 \\ 4 & 2 \\ 6 & 8 \\ 8 & 6 \end{bmatrix}$$

→ compute mean $(\bar{x}, \bar{y})$

$$\bar{x} = \frac{2+4+6+8}{4} = \frac{20}{4} = 5$$

$$\bar{y} = \frac{4+2+8+6}{4} = \frac{20}{4} = 5$$

→ Center the data

$$\begin{bmatrix} 2-5 & 4-5 \\ 4-5 & 2-5 \\ 6-5 & 8-5 \\ 8-5 & 6-5 \end{bmatrix} = \begin{bmatrix} -3 & -1 \\ -1 & -3 \\ 1 & 3 \\ 3 & 1 \end{bmatrix}$$

Pase 2 compute Variance & covariance

$$Var(x) = \frac{(-3)^2 + (-1)^2 + (1)^2 + (3)^2}{4} = \frac{20}{4} = \underline{5}$$

$$Var(y) = \frac{(-1)^2 + (-3)^2 + (3)^2 + (1)^2}{4} = \frac{20}{4} = \underline{5}$$

$$cov(x,y) = \frac{(-3)(-1) + (-1)(-3) + (1)(3) + 3(1)}{4} = \frac{12}{4} = \underline{3}$$

The matrix: $\Sigma = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}$

Pase 3 compute Eigenvalues (λ): $\det(\Sigma - \lambda I) = 0$

$$\begin{vmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = (5-\lambda)^2 - 9 = 0$$

$$\lambda^2 - 10\lambda + 25 - 9 = 0 \Rightarrow \lambda^2 - 10\lambda + 16 = 0$$

solving for λ

$$\underline{\lambda_1 = 8} \text{ and } \underline{\lambda_2 = 2}$$

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Phase 4 Find the direction for the largest eigenvalue ($\lambda_1 = 8$)

$$(\Sigma - 8I) = 0 \Rightarrow \begin{bmatrix} 5-8 & 3 \\ 3 & 5-8 \end{bmatrix} = \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$-3x + 3y = 0$$

$$3x = 3y, \quad \underline{x = y}$$

Eigenvector $(v_1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$(\Sigma - 2I) = 0 \Rightarrow \begin{bmatrix} 5-2 & 3 \\ 3 & 5-2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$3x + 3y = 0$$

$$x = -y$$

Eigenvector $(v_2) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

<u>Phase 5</u> Final 1D transformation

Since λ_1 has larger value I select it as PC_1 (principal component 1)

Normalize Vector $(u) = \frac{1}{\sqrt{1^2 + 1^2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.707 \\ 0.707 \end{pmatrix}$

$$z = x \cdot 0.707 + y \cdot 0.707$$

Point A: $(-3(0.707) + (-1)(0.707)) = -2.828$

Point B: $(-1(0.707) + (-3)(0.707)) = -2.828$

Point C: $(1(0.707) + (3)(0.707)) = 2.828$

Point D: $(3(0.707) + 1(0.707)) = 2.828$